

Qatar-2b

R-1998

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140619 · created 2026-08-07T15:46:15+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456827.672177 +/- 6.1e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.045 +/- 0.056
Transit depth (Rp/Rs)^2	0.21 +/- 0.51 [%]
Orbital Inclination (inc)	84.06 +/- 1.75
Airmass coefficient 1 (a1)	0.94 +/- 0.046
Airmass coefficient 2 (a2)	0.048 +/- 0.095
Scatter in the residuals of the lightcurve fit is	4.93 %
Best Comparison Star	#1 - [192, 75]
Optimal Aperture	2.45
Optimal Annulus	10.05
Transit Duration (day)	0.0209 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.045 $\pm$ 0.056 vs 0.16327 (deviation 2.11 σ)
Duration fit vs published	0.0209 d vs 0.0767 d (deviation 8.45 σ)
Mid-time O-C	0.2 min
Depth SNR	0.8
Residual scatter	4.93 %

Light curve

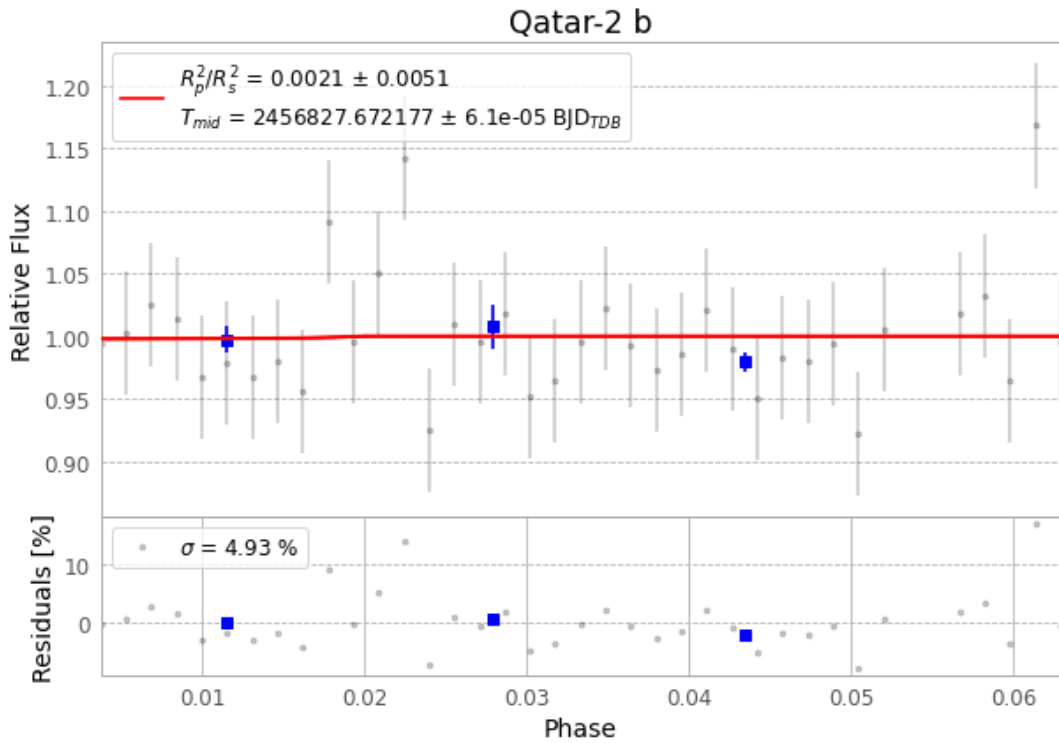


Figure 1 — FinalLightCurve_Qatar-2 b_2014-06-18.png (SHA-256 96c2a7d8a4208d83...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [378, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6df4fe00153fdc6c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

39 FITS frames (26 MB), Qatar-2140619040913.FITS → Qatar-2140619060312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140619.log (482ca48fd596...), FinalLightCurve_Qatar-2 b_2014-06-18.png (96c2a7d8a420...), FinalLightCurve_Qatar-2 b_2014-06-18.csv (75784067b306...)

Compute resources

Wall time 91.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:46Z.